PERFECT DIRECT IMAGE OF THE INTERNAL HOM FROM A PERFECT OBJECT INTO A BOUNDED FLAT FINITELY PRESENTED COMPLEX OF PROPER SUPPORT COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The formation of the derived pushforward of the internal Hom commutes with arbitrary base change under the stated hypotheses. invoked at proof line 7856. Reduces the lemma to proving that K is a perfect object.
- Over a Noetherian base the object Rf_* R SheafHom(E, G^bullet) is perfect, by the Noetherian internal-Hom lemma. invoked at proof lines 7858-7859 and again at proof lines 7884-7886. Supplies the Noetherian case the approximation argument reduces to.
- Noetherian approximation: schemes, bounded finitely presented complexes, their flatness, properness of support, and perfect objects all descend from a filtered colimit of Noetherian rings after enlarging the index, by the Limits descent lemmas. invoked at proof lines 7865-7883. Produces the Noetherian model X_i, G_i^bullet, E_i over R_i to which the Noetherian case applies.

Source reference. Original lemma 08IE: sources/stacks-perfect/pinned-original-source.tex, source lines 6928–6928. Cited at proof line 7856 to dispose of the base change half; the perfection half is inside the counted span.

Source reference. Original lemma 0DJR: sources/stacks-perfect/pinned-original-source.tex, source lines 7063–7063. Cited at proof lines 7858-7859 for the Noetherian case and at proof lines 7884-7886 on the descended datum; the approximation argument is inside the counted span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof lines 7866-7868 to descend X to X_i; the descent chain is inside the counted span.

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Cited at proof lines 7870-7872 to descend G^bullet to G_i^bullet; the descent chain is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05LY: sources/stacks-chow/tags.txt, source lines 7252–7252. Cited at proof lines 7874-7875 to make G_i^n flat over R_i; the descent chain is inside the counted span.

Source reference. Original lemma 081G: sources/stacks-chow/tags.txt, source lines 10216–10216. Cited at proof lines 7876-7878 for properness of the support of G_i^n; the descent chain is inside the counted span.

Source reference. Original lemma 0CYS: sources/stacks-chow/tags.txt, source lines 16252–16252. Cited at proof lines 7879-7880 together with the Limits lemma; the descent chain is inside the counted span.

Source reference. Original lemma 09RF: sources/stacks-perfect/pinned-original-source.tex, source lines 7532–7532. Cited at proof line 7881 to descend E to a perfect E_i; the descent chain is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of finite presentation. Let $E \in D(\mathcal{O}_X)$ be a perfect object. Let $\mathcal{G}^\bullet$ be a bounded complex of finitely presented $\mathcal{O}_X$-modules, flat over S, with support proper over S. Then*

$$K = Rf_* R\mathcal{H}om(E, \mathcal{G}^\bullet)$$

is a perfect object of $D(\mathcal{O}_S)$ and its formation commutes with arbitrary base change.

Proof. The statement on base change is Lemma 08IE. Thus it suffices to show that K is a perfect object. If S is Noetherian, then this follows from Lemma 0DJR. We will reduce to this case by Noetherian approximation. We encourage the reader to skip the rest of this proof.

The question is local on S, hence we may assume S is affine. Say $S = \mathrm{Spec}(R)$. We write $R = \mathrm{colim}\, R_i$ as a filtered colimit of Noetherian rings R_i. By Limits, Lemma 01ZM there exists an i and a scheme X_i of finite presentation over R_i whose base change to R is X. By Limits, Lemma 01ZR we may assume after increasing i, that there exists a bounded complex of finitely presented $\mathcal{O}_{X_i}$-modules $\mathcal{G}_i^\bullet$ whose pullback to X is $\mathcal{G}^\bullet$. After increasing i we may assume $\mathcal{G}_i^n$ is flat over R_i, see Limits, Lemma 05LY. After increasing i we may assume the support of $\mathcal{G}_i^n$ is proper over R_i, see Limits, Lemma 081G and Cohomology of Schemes, Lemma 0CYS. Finally, by Lemma 09RF we may, after increasing i, assume there exists a perfect object E_i of $D(\mathcal{O}_{X_i})$ whose pullback to X is E. Applying Lemma 0DJR to $X_i \to \mathrm{Spec}(R_i)$, E_i, $\mathcal{G}_i^\bullet$ and using the base change property already shown we obtain the result. $\square$